

Jay Bhatt

Edmonton, AB, Canada

♥ Edmonton, AB | ✉ bbaj2404@gmail.com | 🌐 github.com/jay-a-bhatt

SKILLS

Languages: Python, C, C#, Rust, Kotlin, Go, Odin, Bash

Technologies: Linux/Unix, Git, Unraid, Proxmox, Docker, SQL (Microsoft, MySQL, SQLite), Jira

RESEARCH

Undergraduate Research Assistant

Edmonton, AB

Grant MacEwan University

May 2026 - Present

- Building a normative modeling pipeline to analyze structural MRI in ALS patients from the CALSNIC dataset
- Applied pretrained lifespan Bayesian linear regression models to FreeSurfer cortical thickness and subcortical volume features, learning site specific offsets via adaptation to compute per region Z scores and centiles
- Quantified patient-vs-control deviations with Cohen's d and Welch t-tests under Benjamini-Hochberg FDR correction, and validated model calibration on held-out healthy controls

EXPERIENCE

Pharmacy Assistant

Edmonton, AB

Ellerslie Pharmacy

March 2025 - Present

- Ensured strict regulatory compliance in prescription processing, packaging, and labeling; direct experience with data accuracy and governance standards
- Maintained patient records in pharmacy management software while upholding data integrity and privacy requirements
- Identified and resolved technical risks in service computers, reducing potential data loss and operational downtime

Lunchtime Supervisor

Edmonton, AB

Jan Reimer School

April 2022 - Present

- Maintained safe and documented environments; escalated incidents through appropriate reporting chains

PROJECTS

Ancient DNA Classifier

Private Repo

- Built and evaluated traditional machine learning and deep learning models to classify human mitochondrial DNA as ancient or modern
- Implemented a suite of ML algorithms, including Logistic Regression, Random Forest, Gaussian Naive Bayes, SVM, KNN, XGBoost, and CatBoost.
- Designed and trained a deep learning model using PyTorch

F1 Lap Time Prediction

Private Repo

- Developed an end-to-end machine learning pipeline to predict Formula 1 lap times, managing feature engineering and model evaluation across the full ML lifecycle.

EmBlocks

github.com/IsaacMcCracken/EmBlocks-HackED-2026

- Developed a block-based IDE and C-level abstraction layer for the RP2040 microcontroller, enabling simplified embedded system development through Scratch-like visual programming.

Fluid Simulation

github.com/jay-a-bhatt/Fluid-Simulation-hackED-2025

- Developed a high-performance fluid simulation using Rust and WebGPU, implementing PIC/FLIP algorithms for realistic, real-time particle dynamics.
- Optimized for cross-platform web deployment using WebAssembly (WASM), featuring a responsive UI with custom controls for interactive user manipulation.

Home Server

- Self-host and maintain a 15+ service environment (Unraid, Proxmox, Docker), managing secure access and control
- Hardened infrastructure using Private Key SSH, CrowdSec IPS, and MFA-backed SSO (Authentik) to enforce strict identity-aware access control.

EDUCATION

Grant MacEwan University, Edmonton, AB

2026 Expected

BSc Computer Science Major + Biology Minor

- **CMPT Courses:** Software Engineering, Machine Learning, Robotics, Databases, Networking, Cryptography, Operating Systems, Human/Computer Interaction, Data Structures & Algorithms, Numerical Methods
- **BIOL Courses:** Eukaryotic Cell Biology, Genetics, Evolution, Ecology

W.P. Wagner High School, Edmonton, AB

June 2021

High School Diploma